

CLASS XII · LECTURE 1

The Shape of a C Program

Option I · Two periods · Structure, #include, #define, reserved words

By the end of this lecture you can

- Write the skeleton of a C program that prints one line.
- Explain #include and #define with an example of each.
- State the rules of an identifier and name reserved words.
- Distinguish scanf from getchar.

What you are being asked to write

C is a high-level language designed by Dennis Ritchie. A compiler translates your source program into object code the processor can run. Every complete program in this option has a function called **main**. The machine starts there. A program with no main does not have a starting point, which is why the multiple-choice line “the only function all C programs must contain” is **main**.

```
#include <stdio.h>

int main(void) {
    printf("Computer Science\n");
    return 0;
}
```

Program 1.1 The smallest honest program. It prints one line and ends cleanly.

Read it from the top. `#include <stdio.h>` asks the preprocessor to bring in the standard input/output header, which declares `printf` and `scanf`. The line `int main(void)` starts the main function and promises that it returns an integer and takes no arguments. `printf` sends text to the screen. `\n` inside the quotes moves to the next line. `return 0` tells the operating system that the program finished normally. Comments, which the compiler ignores, sit between `/*` and `*/`.

EXAM TIP Turbo C in the college lab

Some labs still use Turbo C and write void main, clrscr and getch from conio.h. The board marks the logic. Prefer the form above when you are writing on the answer script: `stdio.h`, `int main`, and `return 0`. If your teacher has drilled the Turbo C form, either is accepted when the program is otherwise correct. Do not mix `clrscr` into a program and then forget `stdio.h`.

#include and #define

Both lines are **preprocessor directives**. They are handled before compilation. They have no semicolon.

Directive	Job	Example
#include	Inserts a header so you can call its functions	#include <stdio.h>
#define	Gives a name to a constant or a short macro. The name is replaced by the value before compile time	#define PI 3.1416

```
#include <stdio.h>
#define PI 3.1416

int main(void) {
    float r = 7;
    printf("Area = %f\n", PI * r * r);
    return 0;
}
```

Program 1.2 #define used for a constant. There is no equals sign and no semicolon on the define line.

ON THE BOARD Two examples, as the question asks

#include <stdio.h> brings in printf and scanf. #include <math.h> brings in fabs and sqrt. #define MAX 100 replaces every MAX with 100. #define PI 3.1416 replaces PI with the value. That is four legal examples. Two are enough.

Names, reserved words and comments

An **identifier** is a name you invent for a variable, a constant or a function. It may contain letters, digits and underscores. It must start with a letter or an underscore. It must not be a reserved word. `Number`, `NUMBER5` and `Number_5` are legal. `5number` is not, because it starts with a digit. C treats `Marks` and `marks` as different names.

A **reserved word** already has a meaning in the language, so you cannot use it as a variable name. Learn a working list: `int`, `float`, `char`, `double`, `void`, `if`, `else`, `for`, `while`, `do`, `switch`, `case`, `break`, `continue`, `return`, `struct`. The question “name any three” is that list. Do not invent words such as `print` or `loop` and call them reserved.

DEFINITION scanf and getchar

`scanf` reads formatted input. You give a format and the address of the variable: `scanf("%d", &n)` reads an integer into `n`. `getchar` reads exactly one character and returns it. `scanf` can skip spaces and read a whole number. `getchar` does not format the input; the next character, which might be a newline left behind, is what you get.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. Every complete C program must contain:

- A) `start()`
- B) `getch()`
- C) `main()`
- D) `clrscr()`

2. Which identifier is illegal?

- A) `Number`
- B) `5number`
- C) `NUMBER5`
- D) `Number_5`

3. C was designed by:

- A) Dennis Ritchie
- B) Charles Babbage
- C) Tim Berners-Lee
- D) Bill Gates

Short answers

S1. Write the basic structure of a C program.

S2. Define `#include` and `#define`. Give one example of each.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. C. Execution begins at main.

MCQ 2. B. An identifier cannot start with a digit.

MCQ 3. A. Ritchie designed C. Babbage belongs with early mechanical computing.

S1. A C program includes the headers it needs, then defines main. Execution starts at main. A minimal program is: `#include <stdio.h>` then `int main(void) { printf("Hello\n"); return 0; }`. The include line has no semicolon. `return 0` ends main successfully.

S2. `#include` is a preprocessor directive that inserts a header file. Example: `#include <stdio.h>` provides `printf` and `scanf`. `#define` gives a name to a value that is substituted before compilation. Example: `#define PI 3.1416`. Neither directive ends with a semicolon.